MEHDI BELHAJ

Embedded Systems Engineer · FPGA & Software

Digital and mixed-signal boards · Safety-critical embedded software · Digital control · Aerospace

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`FPGA · VHDL · NIOS II` `MCU C2000` `C · C++ · Python` `UART · SPI · I2C · RS422`



AVAILABILITY	EXPERIENCE	CONTRACT	MOBILITY	WORK PERMIT	DRIVING LICENCE	ENGLISH
Immediate	3 years	Permanent · Fixed-term · Umbrella	All of France	Valid to Nov. 2028	B	C1 · fluent

PROFILE

Embedded systems engineer (MSc in EEA) with 3 years of experience, including safety-critical aerospace equipment (Safran, Airbus): microcontroller-based control boards, interfaces and measurement chains, contribution to **DAL A software**. Trained in **FPGA** design and certified in **C++ and Python**, I bridge hardware, software and control, from specification to test plans.

SKILLS

FPGA & DIGITAL	**FPGA** Altera Cyclone IV: **VHDL** (synchronous logic, PWM), Quartus, Qsys, **NIOS II** soft-core · **TMS320 C2000** and PIC microcontrollers: PWM, ADC, interrupts
EMBEDDED SOFTWARE	**C** (interrupts, low-level software) · **C++17** (OOP, STL) · **Python** · PyTest and GoogleTest, continuous integration · Git, Linux
INTERFACES & BOARDS	**UART, SPI, I2C, RS422** buses · Data acquisition (24-bit ADC), current, voltage, temperature and force sensors · **OrCAD Capture CIS**, KiCad · Power supplies and protections
DIGITAL CONTROL	**MATLAB / Simulink / Simscape** · Discretised PI / PID controllers · BLDC and induction motor control · PSIM
TESTING & QUALIFICATION	HW and EMC test plans, automated test bench instruction · Oscilloscope, LabVIEW · DO-160G (Sections 16, 21, 22, 25) · DAL A · CE and UKCA marking (EN 60335, EMC, RED)

PROFESSIONAL EXPERIENCE

Hardware Electronics Engineer — Safran Ventilation Systems

`Nov. 2023 – May 2025 · 19 months`

Via Centum T&S · Blagnac, France · EMC DO-160G category H

1,890 W input board of a 270 VDC aircraft fan mock-up, interfaced with the motor control.

- **Interfaces with the control**: phase-loss signal to the control board in under 4 ms, temperature measurement for the software, power rails up in under 200 ms and held for 200 ms during supply interruptions.
- **Power supplies**: multi-output isolated Flyback (132 kHz) from 270 VDC, 5, 16 and 18 V rails via LDOs and a 1.5 MHz Buck.
- **Validation and deliverables**: simulation, HW and EMC test plans, automated test bench instruction (ITE); DO-160G design justification file (DJD), layout specification; weekly meeting in English with Safran USA.

Tools: PSIM · OrCAD Capture CIS · Excel (calculation notes) · Windchill PLM

Results: 270 VDC feasibility demonstrated with minimal impact on software and mechanics.

Electronics & Embedded Systems Engineer, R&D — Airbus / Crouzet

`Aug. 2022 – Nov. 2023 · 16 months`

Via Serma Ingénierie · Toulouse · Zero Emission programme · ABD0100 requirements

Control electronics of an electromechanical spoiler actuator (deployed in under 130 ms): 3 stacked boards (target volume 65 × 65 × 170 mm), DAL A.

- **CPU board**: TMS320F280049 (C2000) microcontroller — pin assignment with the software team (PWM, ADC, serial buses, watchdog); supplies from the 28 V bus (3.3 V Flyback, 20 ms energy reserve); force (24-bit ADC ADS1247) and position measurement.
- **DAL A low-level software**: contribution to the equipment software (EQSW), the sequencer and the BITE self-tests, supporting the Airbus application.
- **Control**: Simulink model of the BLDC motor and its control (Hall sensors, position / current loops), mechatronic modelling, PI tuning.
- **Specification and safety**: electronic sub-assembly specifications, hardware and software requirements with the Safety team.

Tools: MATLAB / Simulink / Simscape · Scilab / Xcos · PSIM · LTspice · OrCAD · KiCad

Results: 130 ms requirement validated in simulation; architecture approved in Airbus and Crouzet design reviews.

Final-year internship, R&D power electronics — SuperGrid Institute — Feb. – Jul. 2022 · 6 months
Villeurbanne, France · research institute for HVDC grids

Isolated supply for the gate drivers of 6 series-connected 1.7 kV SiC MOSFETs — 12 transformers, insulation validated at 20 kV.
Tools: LTspice · oscilloscope

FPGA & SOFTWARE PROJECTS

- **C99 and Python — bldc-control-sim** (2026): BLDC motor control from the model to the embedded code — C99 controller for a Cortex-M4F (field-oriented and six-step control, no dynamic allocation, 3.1 kB), tested in simulation against its Python reference on an inverter with dead time; 971 C checks, 45 Python tests. **Tools**: gcc · clang · GCC ARM · PyTest · GitHub Actions
- **FPGA — BLDC motor control** (ENSEEIHT, 2021): commutation and PWM modelled, PWM and speed measurement coded in VHDL on a DE0-Nano board (Cyclone IV), NIOS II processor, control law discretised and coded in C with interrupts; speed control validated on a real motor. **Tools**: PSIM · MATLAB · Quartus · Qsys
- **C++17 — embedded-dsp-cpp**: library for embedded targets — heap-free ring buffer, moving average, IIR low-pass filter; 12 unit tests. **Tools**: CMake · GoogleTest · GitHub Actions
- **Python — hardware-design-kit** (2026): power-side design calculators for a board — input protections (precharge, DO-160G lightning, ESD), EMI filter, flyback, measurement chains and error budgets, PCB rules (IPC-2221, IEC 60664-1); 76 tests, ngspice cross-checks. **Tools**: NumPy · SciPy · ngspice · PyTest · GitHub Actions
- **Python — scope-emi-toolkit**: EMC test automation — parsing of oscilloscope CSV exports (Tektronix, Rigol), windowed spectrum calibrated in dBµV, comparison with a 150 kHz – 152 MHz conducted-emissions mask and pass / fail report; 8 unit tests, continuous integration. **Tools**: NumPy · SciPy · Matplotlib · PyTest · GitHub Actions

ENTREPRENEURSHIP

Co-founder — MANMAFOX SAS — Since Jul. 2023
TokyoSeat smart toilet seats · Amazon EU · side activity

- Product compliance with the manufacturer and an ISO/IEC 17025-accredited test laboratory (CNAS): EN 60335, EMC, radio and RoHS tests, test-report analysis, CE and UKCA marking, WEEE, technical files.

EDUCATION & CERTIFICATIONS

Master's degree in EEA — Power Electronics, Actuators & Control · Université Toulouse III Paul Sabatier · INP-ENSEEIHT — 2020 – 2022
Projects: induction machine control (Saber), current-mode switch-mode power supply

Bachelor's degree in EEA · Université Toulouse III Paul Sabatier — 2019

Two-year technical degree (DUT) in Electrical Engineering and Industrial Computing · ENSAM Rabat, Morocco — 2018

C++ and Python certifications — Cisco Networking Academy / OpenEDG — 2025 – 2026
C++ Essentials 1 and 2, C++ Advanced, Python Essentials 1 and 2 · 330 h



French electrical safety authorisation B0, H0, B2V, B2V Essai, BE Essai — CAMIRA Formation — 2022 – 2025

EYE Berlin — European Youth Event, European Parliament · certificate of participation — Apr. 2024

LANGUAGES

French bilingual · **English** C1, professional use · **Spanish, Portuguese** basic

SOFT SKILLS & INTERESTS

Methodical · autonomous · rigorous · curious · team player · Competitive football · NQT mentoring programme · Travel (18 countries)